Scaling Book Exercises: Section 7

Lucas Yan

Chapter: All About Transformer Inference
Source scan: `../handwritten/section_07_handwritten.pdf`
Reference: `https://jax-ml.github.io/scaling-book/inference/`

Model. The worked problems use

$$L = 64, \quad D = 4096, \quad F = 16384, \quad N = 32, \quad K = 8, \quad H = 256, \quad V = 32128.$$

Here L is the number of layers, D is the model dimension, F is the feed-forward dimension, N is the number of query heads, K is the number of KV heads, H is the head dimension, and V is the vocabulary size.

Exercise 1

The gated MLP has three $D \times F$ matrices per layer, so

$$P_{\text{MLP}} = 3LDF = 3(64)(4096)(16384) = 12{,}884{,}901{,}888.$$

The attention matrices contribute

$$\begin{aligned} P_{\text{attention}} &= L\big(DNH + DKH + DKH + DNH\big) \\ &= 2LDH(N+K) \\ &= 2(64)(4096)(256)(32+8) \\ &= 5{,}368{,}709{,}120. \end{aligned}$$

Because the input and output embedding matrices are shared,

$$P_{\text{vocab}} = DV = (4096)(32128) = 131{,}596{,}288.$$

Therefore,

$$\begin{aligned} P &= 3LDF + 2LDH(N+K) + DV \\ &= 18{,}385{,}207{,}296 \\ &\approx \boxed{18.4\text{B parameters}}. \end{aligned}$$

For each token, every layer caches one key and one value for each KV head. Thus the KV cache contains

$$2LKH = 2(64)(8)(256) = 262{,}144$$

elements per token. With int8 elements,

$$\boxed{262{,}144 \text{ bytes/token} \approx 262 \text{ kB/token} = 256 \text{ KiB/token}}.$$

Exercise 2

A TPU v5e 4×4 slice contains 16 chips, each with 16 GB of HBM:

$$M_{\text{slice}} = 16(16 \text{ GB}) = 256 \text{ GB}.$$

The int8 model weights occupy approximately 18.4 GB. A 128,000-token KV cache for one sequence occupies

$$(262{,}144 \text{ bytes/token})(128{,}000 \text{ tokens}) = 33.55 \text{ GB}.$$

The maximum batch size is therefore

$$B_{\max} = \left\lfloor \frac{256 - 18.4}{33.55} \right\rfloor = \boxed{7}.$$

If K falls from 8 to 1, each KV cache is eight times smaller, so

$$\boxed{B_{\max} \approx 8(7) = 56}.$$

Exercise 3

The ideal parameter-loading time is bytes divided by aggregate HBM bandwidth. Each TPU v5e supplies approximately 8.2×10^{11} bytes/s, so the 4×4 slice supplies

$$16(8.2 \times 10^{11}) = 1.312 \times 10^{13} \text{ bytes/s}.$$

For 18.4 GB of int8 parameters,

$$T_{\text{parameters}} = \frac{18.4 \times 10^9}{1.312 \times 10^{13}} \approx 1.4 \times 10^{-3} \text{ s} = \boxed{1.4 \text{ ms}}.$$

This is an optimistic lower bound assuming perfect sharding and full HBM bandwidth utilization.

Exercise 4

Sharding

A TPU v5e 4×4 slice has two physical ICI dimensions, each four chips wide. For prefill, the approximate tensor-parallel roofline is

$$p_{\text{prefill}} \approx \frac{F}{2200} = \frac{16384}{2200} \approx 7.4.$$

A natural mapping therefore uses one four-chip axis for tensor parallelism and the other for sequence parallelism.

Decode is bandwidth-bound, so tensor parallelism can extend further. Let

$$\beta = \frac{W_{\text{HBM}}}{W_{\text{ICI}}} \approx 8.$$

Comparing parameter-loading time with ICI communication gives

$$p_{\text{decode}} \lesssim \frac{F}{B\beta}.$$

For $B = 7$,

$$\frac{F}{B\beta} = \frac{16384}{7(8)} \approx 293,$$

which is much larger than the available 16 chips. Thus a 16-way model-parallel decode is below this idealized bandwidth roofline.

The KV cache has shape

$$\mathrm{KV}[2, B, S, K, H].$$

Shard KV heads first and batch second:

$$\mathrm{KV}[2, B_Z, S, K_Y, H].$$

One natural 4×4 layout uses $Y = 4$ and $Z = 4$, distributing the KV cache across all 16 chips. This layout requires two AllToAll operations per attention layer to move query and output activations between model-sharded and batch-sharded layouts.

Generation latency

For $B = 7$ and a 128,000-token context, the KV-cache loading time is

$$T_{\mathrm{KV}} = \frac{7(33.55 \times 10^9)}{16(8.2 \times 10^{11})} \approx 17.9 \text{ ms}.$$

The parameter-loading time is 1.4 ms. Using approximately 3.93×10^{14} int8 operations/s per chip, the compute time is

$$T_{\mathrm{compute}} = \frac{2BP}{16C} = \frac{2(7)(18.4 \times 10^9)}{16(3.93 \times 10^{14})} \approx 0.041 \text{ ms}.$$

Parameter loading dominates the linear layers, so the theoretical step time is

$$\begin{aligned} T_{\mathrm{step}} &= T_{\mathrm{KV}} + \max(T_{\mathrm{parameters}}, T_{\mathrm{compute}}) \\ &\approx 17.9 + 1.4 \\ &= \boxed{19.3 \text{ ms}}. \end{aligned}$$

This excludes fixed collective latency and other runtime overheads.

Exercise 5

Suppose the MLP becomes a mixture of experts with $E = 16$ total experts and $k = 2$ activated experts per token.

Parameters

The total parameter count is

$$\begin{aligned} P_{\mathrm{total}} &= 3ELDF + 2LDH(N + K) + DV \\ &= 211{,}658{,}735{,}616 \\ &\approx \boxed{212\mathrm{B}}. \end{aligned}$$

Only $k = 2$ experts are used by a given token, so the activated parameter count is

$$\begin{aligned} P_{\text{activated}} &= 3kLDF + 2LDH(N+K) + DV \\ &= 31{,}270{,}109{,}184 \\ &\approx \boxed{31.2\text{B}}. \end{aligned}$$

Critical batch size

The MoE stores E times as many MLP weights but performs only k times as much MLP computation per token. Therefore its critical batch size increases by approximately E/k. Starting from the bf16 TPU v5e dense-model estimate of 240 tokens,

$$B_{\text{crit}} \approx 240\frac{E}{k} = 240\frac{16}{2} = \boxed{1920 \text{ tokens}}.$$

KV cache and forward FLOPs

MoE changes the MLP, not attention, so the KV cache remains

$$\boxed{2LKH = 262{,}144 \text{ elements/token}},$$

or approximately 262 kB/token in int8.

For T tokens, the forward-pass FLOP count is approximately

$$2P_{\text{activated}}T \approx 2(31.2 \times 10^9)T = \boxed{62.4 \times 10^9 T \text{ FLOPs}}.$$